AI for Multimodal Science 2026 @ University of California, San Diego

Bridging the Lithological Gap: A Foundation-Model Approach to Groundwater Assessment in Northern Nigeria

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Outline of Presentation

#1 Thematic GIS Layers

#2 AHP Model

#3 ML model & its Ensembles

#4 Galileo Multi-Model

#5 Comparison and Results

#6 Q & A

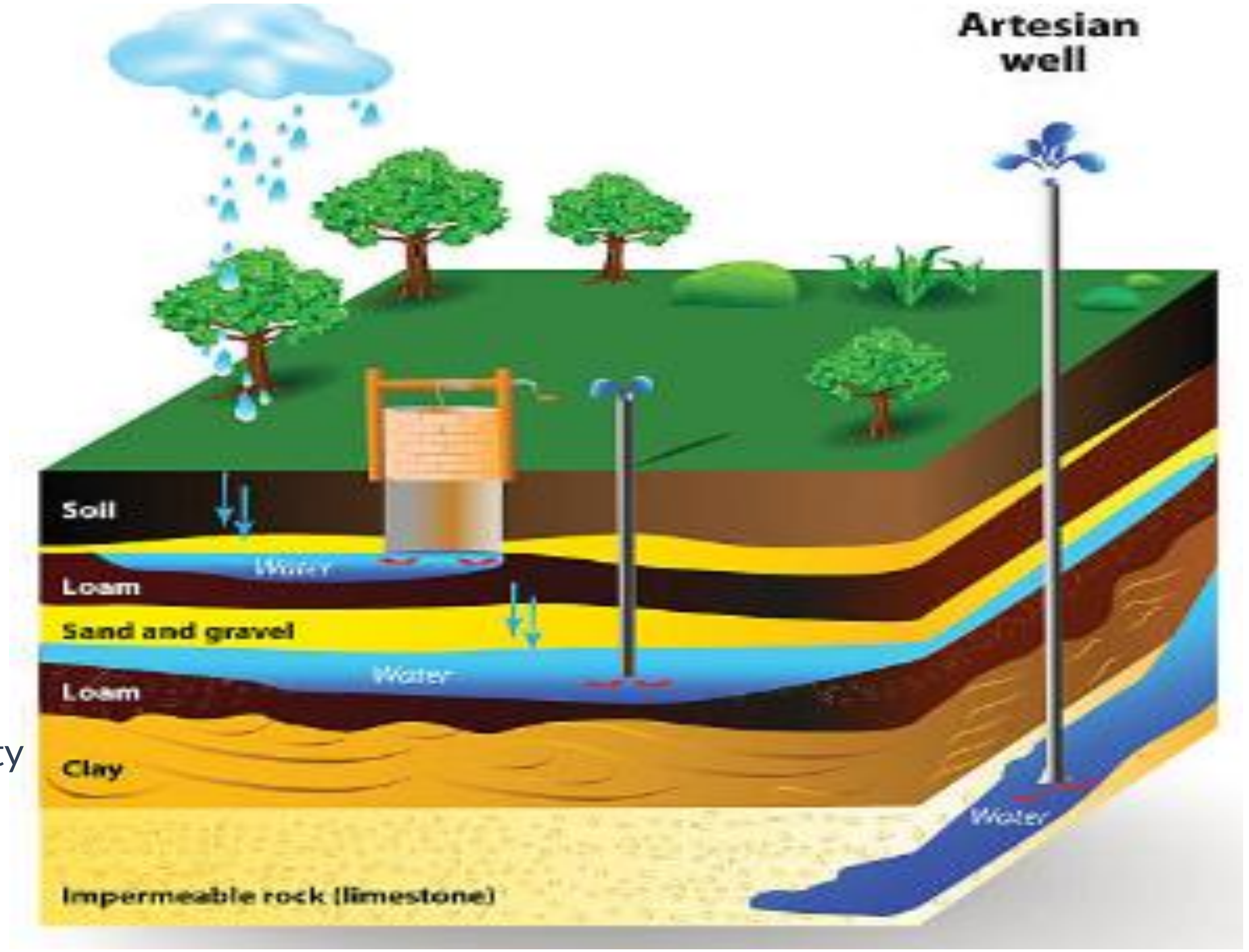
RESEARCH VISION & APPROACH

Challenge: Climate variability and poor resource management threaten water security in Nigeria.

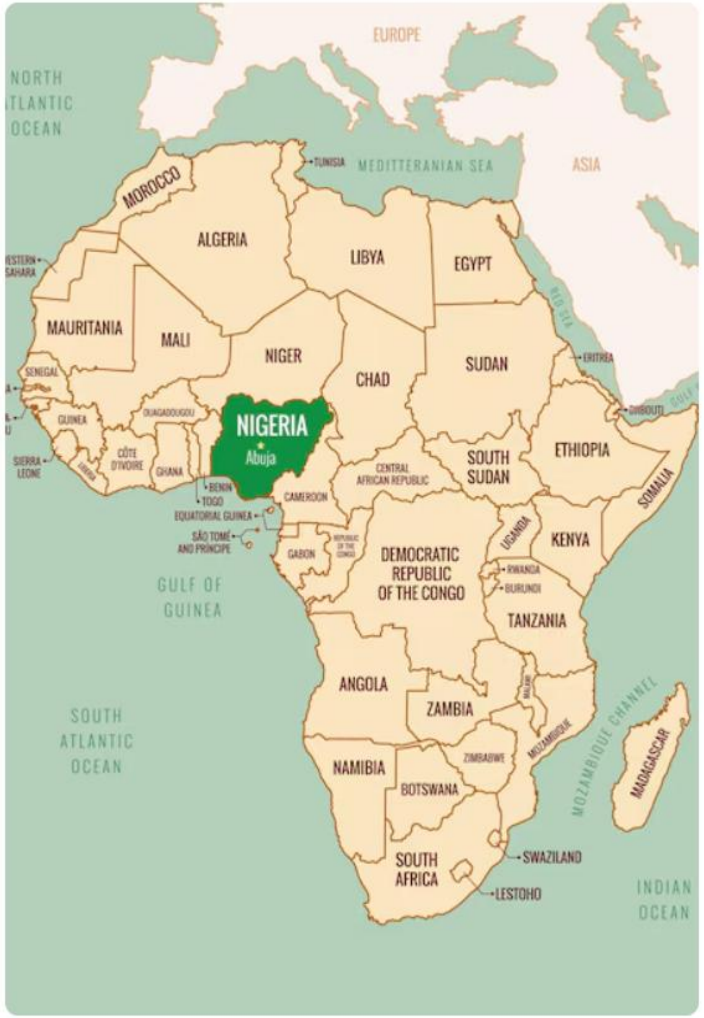
Integration: Combines Geophysics, Remote Sensing, and Machine Learning (ML) for robust analysis.

Goal: Develop an AI-powered predictive model to identify groundwater-favourable zones.

Outcome: A scalable tool guiding groundwater sustainability plan and policy.

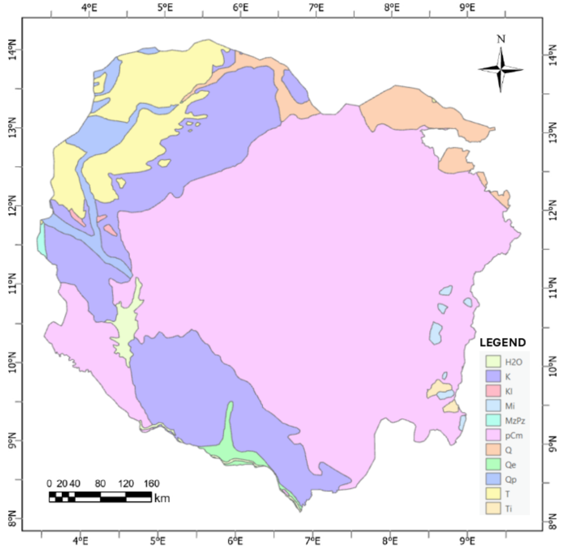


RESEARCH AREA OF INTEREST



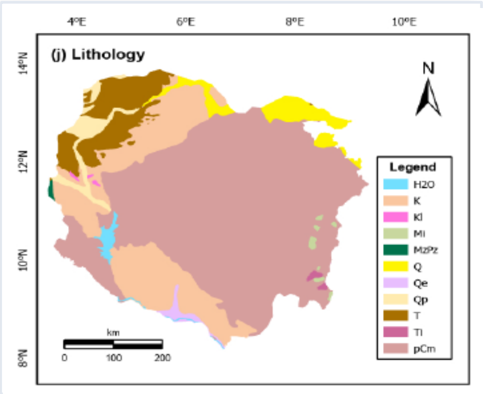
African Map

Nigerian Map

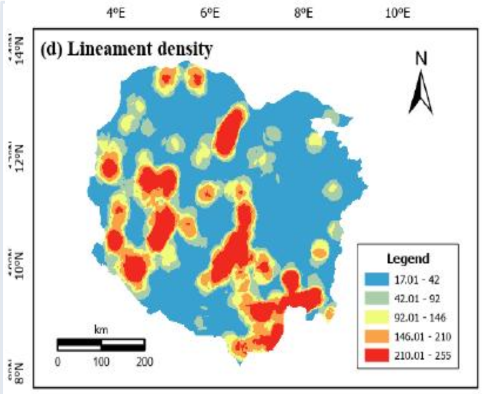


Kaduna · Kano · Katsina · Zamfara · Sokoto · Kebbi · Niger/FCT

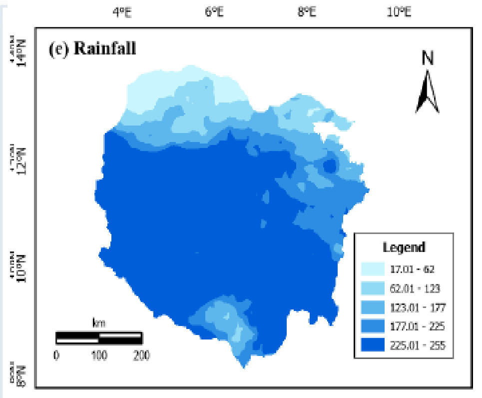
THEMATIC LAYERS (NW NIGERIA) — 10 GIS INPUT PARAMETERS



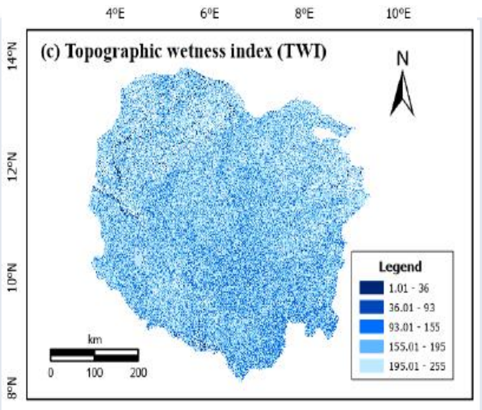
Lithology



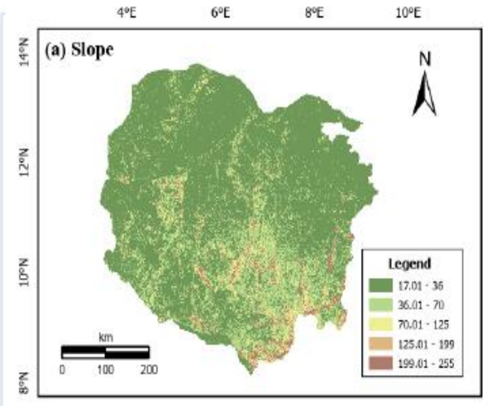
Lineament Density



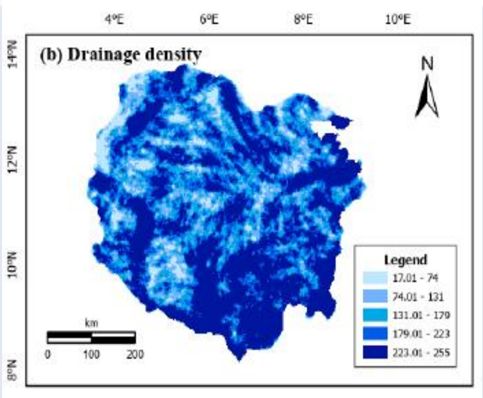
Rainfall



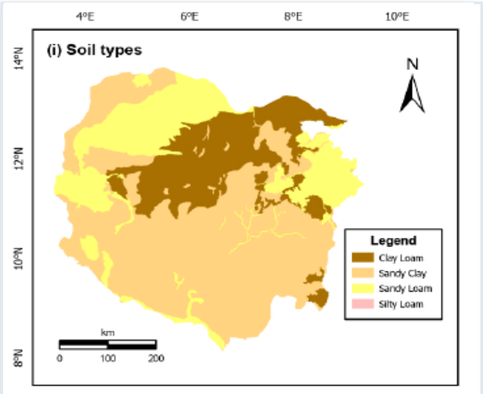
TWI



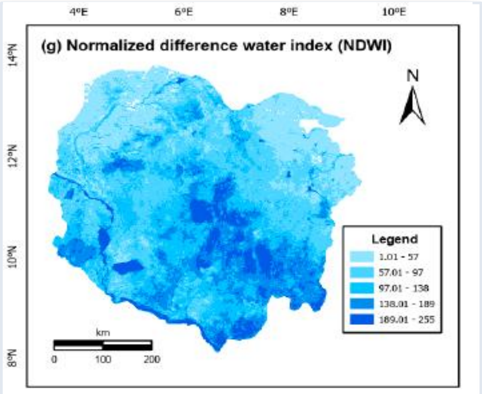
Slope



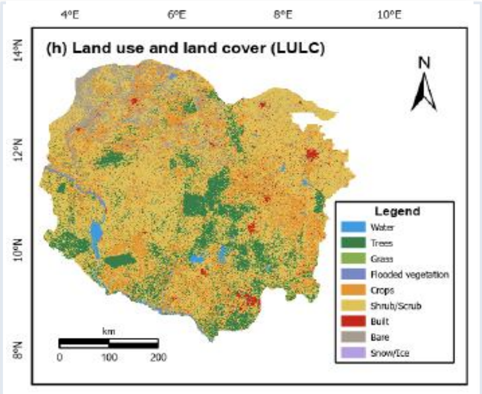
Drainage Density



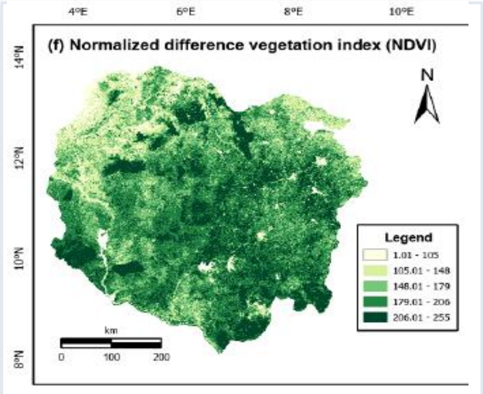
Soil Type



NDWI



LULC



NDVI

Ten thematic remote-sensing and GIS layers used at common resolution

GIS PROCESSING WORKFLOW — ArcGIS Pro

1

Data Acquisition

Sentinel-2 imagery (NDVI, NDWI, LULC)
COPDEM 30m (Slope, TWI, Drainage)

USGS Lithology & Soil datasets

IMERG Rainfall data

Lineament extraction from DEM

2

Coordinate System

Project all rasters to

WGS 1984 UTM Zone 32N

Uniform spatial reference

for all 10 thematic layers

3

Reclassification

Reclassify each layer into

9 groundwater suitability classes

(1=Very Low → 9=Very High)

based on domain knowledge

4

AHP Weighting

Apply Saaty pairwise weights

to each reclassified raster

CR < 0.10 verified

Priority vector normalised

5

Weighted Overlay

Combine all 10 weighted layers

using Raster Calculator:

GWPZ = $\Sigma(Wi \times Ri)$

Output: GWP raster map

6

GWP Zones

Classify final index into:

Very High / High / Moderate

Low / Very Low potential

Export map for validation

Priorities

These are the resulting weights for the criteria based on your pairwise comparisons:

Cat		Priority	Rank	(+)	(-)
1	Lithology	29.4%	1	10.9%	10.9%
2	Lineament	21.5%	2	6.8%	6.8%
3	Rainfall	15.4%	3	4.5%	4.5%
4	Drainage	10.9%	4	3.2%	3.2%
5	Slope	7.6%	5	2.3%	2.3%
6	TWI	5.4%	6	1.7%	1.7%
7	Soil	3.8%	7	1.2%	1.2%
8	LULC	2.7%	8	0.9%	0.9%
9	NDVI	1.9%	9	0.7%	0.7%
10	NDWI	1.5%	10	0.7%	0.7%

Number of comparisons = 45
Consistency Ratio CR = 3.2%

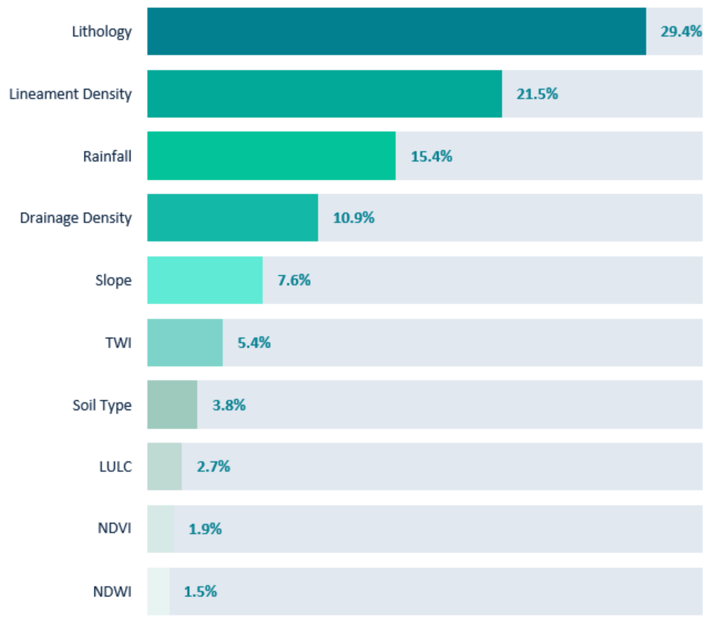
Decision Matrix

The resulting weights are based on the principal eigenvector of the decision matrix:

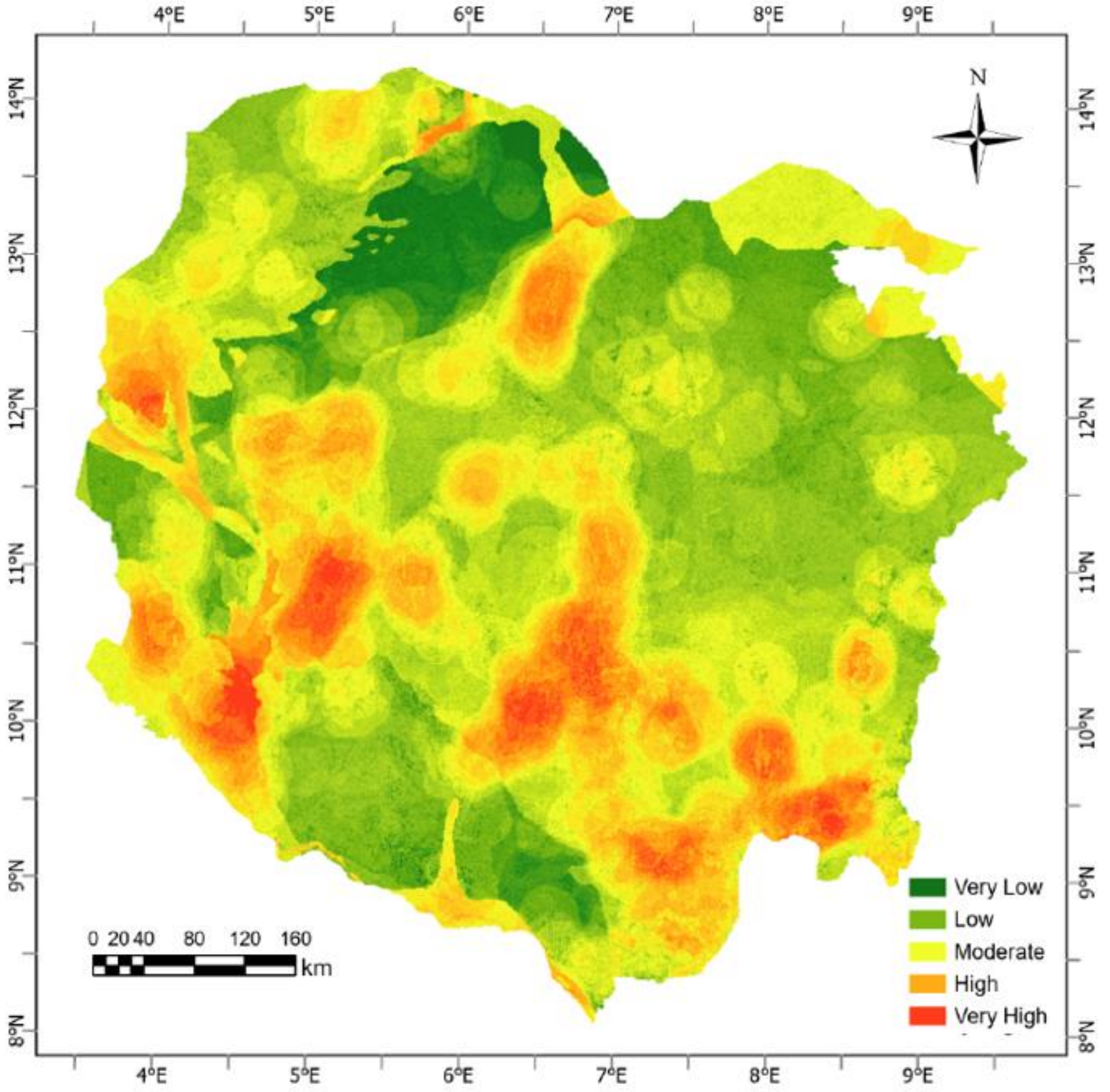
	1	2	3	4	5	6	7	8	9	10
1	1	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00	9.00
2	0.50	1	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00
3	0.33	0.50	1	2.00	3.00	4.00	5.00	6.00	7.00	8.00
4	0.25	0.33	0.50	1	2.00	3.00	4.00	5.00	6.00	7.00
5	0.20	0.25	0.33	0.50	1	2.00	3.00	4.00	5.00	6.00
6	0.17	0.20	0.25	0.33	0.50	1	2.00	3.00	4.00	5.00
7	0.14	0.17	0.20	0.25	0.33	0.50	1	2.00	3.00	4.00
8	0.12	0.14	0.17	0.20	0.25	0.33	0.50	1	2.00	3.00
9	0.11	0.12	0.14	0.17	0.20	0.25	0.33	0.50	1	2.00
10	0.11	0.11	0.12	0.14	0.17	0.20	0.25	0.33	0.50	1

Principal eigen value = 10.424
Eigenvector solution: 5 iterations, delta = 6.8E-8

AHP Criteria Weights (Saaty Pairwise Comparison)



RESULTS — AHP GROUNDWATER POTENTIAL MAP



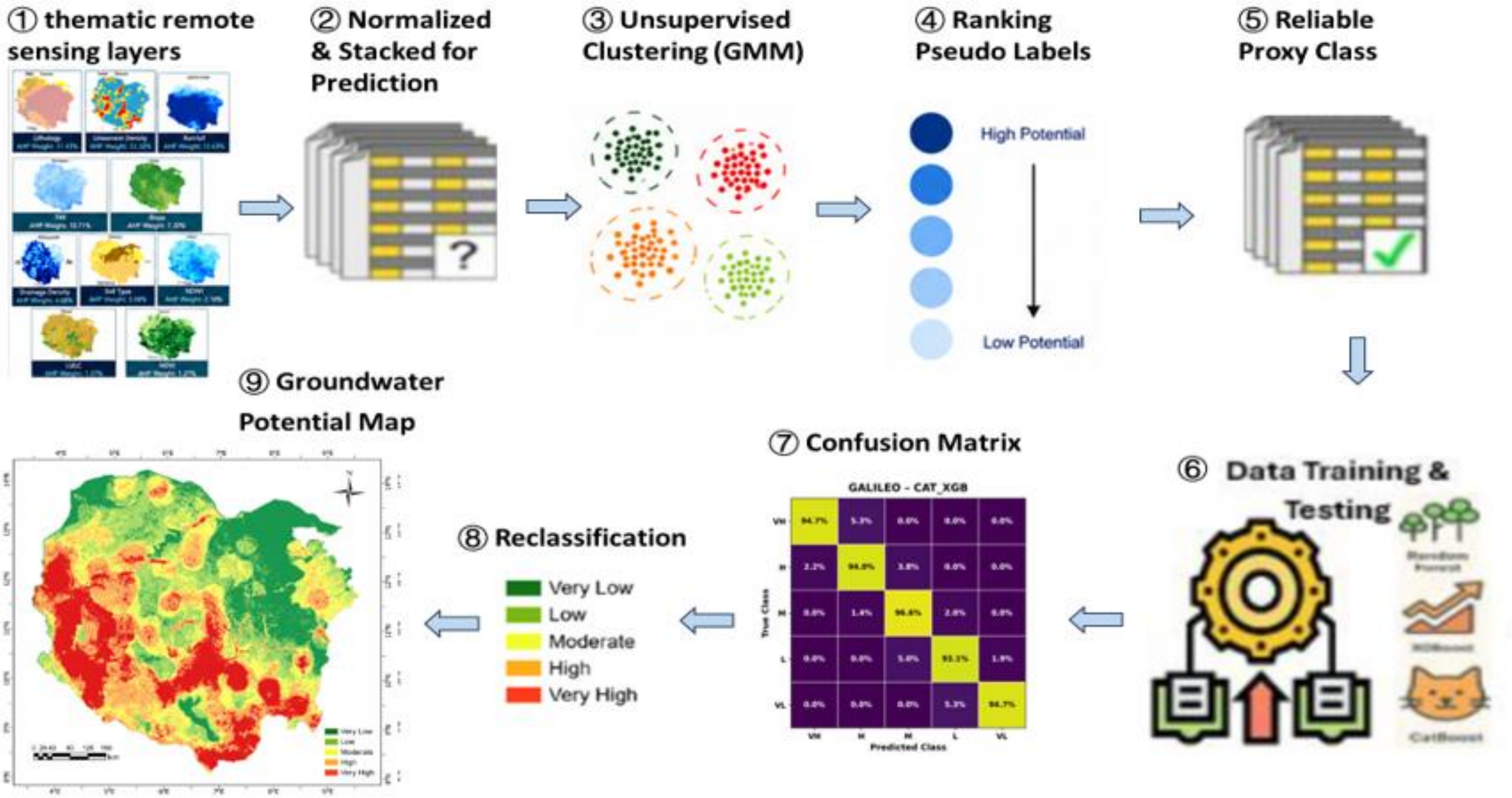
AHP Result Explained

- ✓ CR = 0.032 – Strong reliability
- ✓ Approximately 45-60% AOI is Low to Very Low (Over the Basement Complex)
- ✓ The High to Very High at SW and NW (Sokoto Basin) Favorable for Hydrogeological conditions
- ✓ High zones priority Target for GW development
- ✓ Low zones could be managed for aquifer recharged interventions
- ✓ Result aligns with regional geology with reliable basis for plannings obvious hydrogeological boundaries

Facts

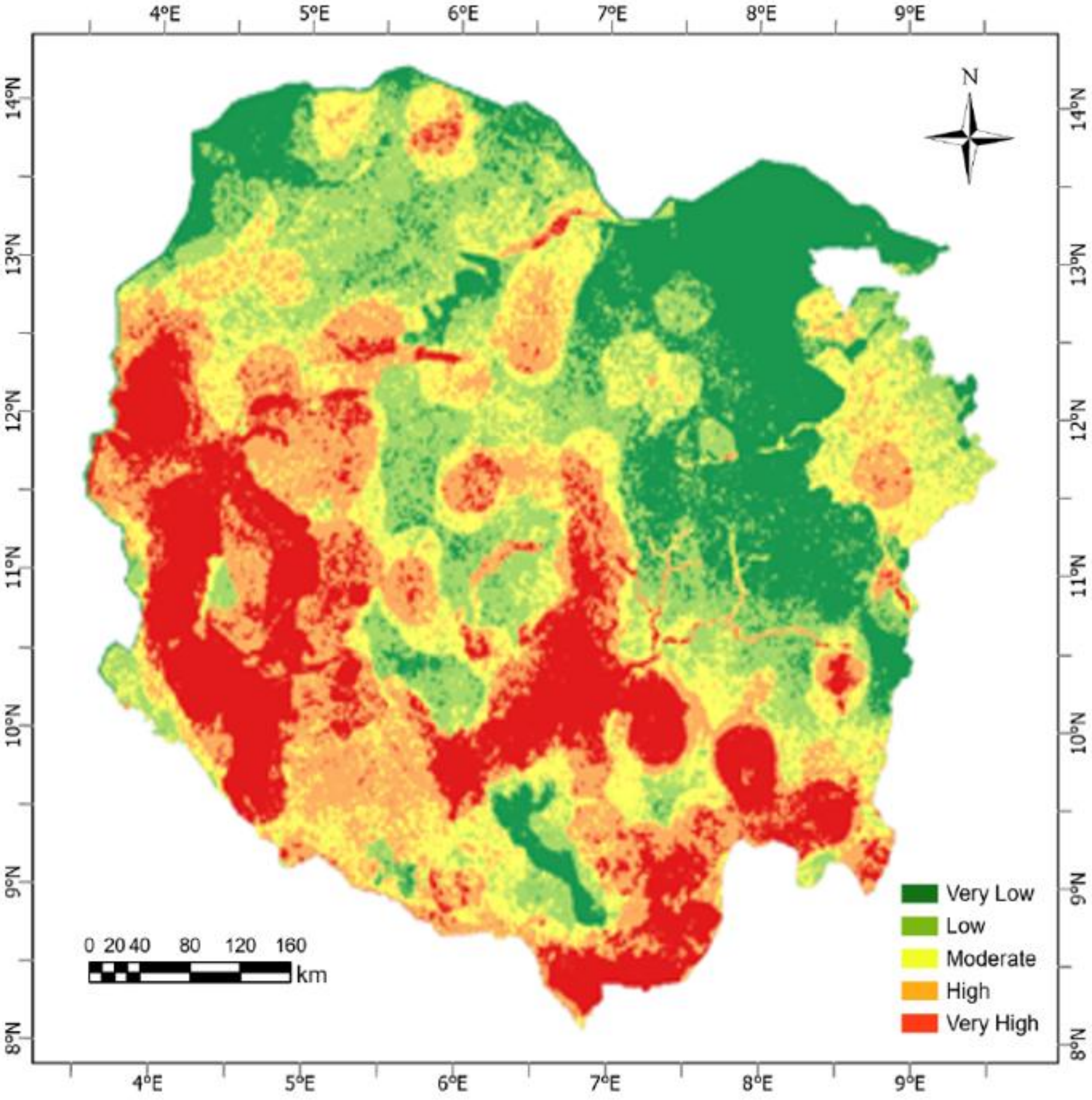
Thematic Layers = 10
CR = 0.032
No Ground Truth Yet

ML MULTI-MODAL FRAMEWORK



Error Metric: Acc:0.954, F1:0.955, Prec:0.955, Rec:0.954, Kappa:0.943

RESULTS —ML (CAT_XGB) GROUNDWATER POTENTIAL MAP



ML Advantages

✓ It has more intense/extensive red (Very High) areas, especially in the southwest and central zones

✓ Shows sharper, more fragmented high-potential clusters

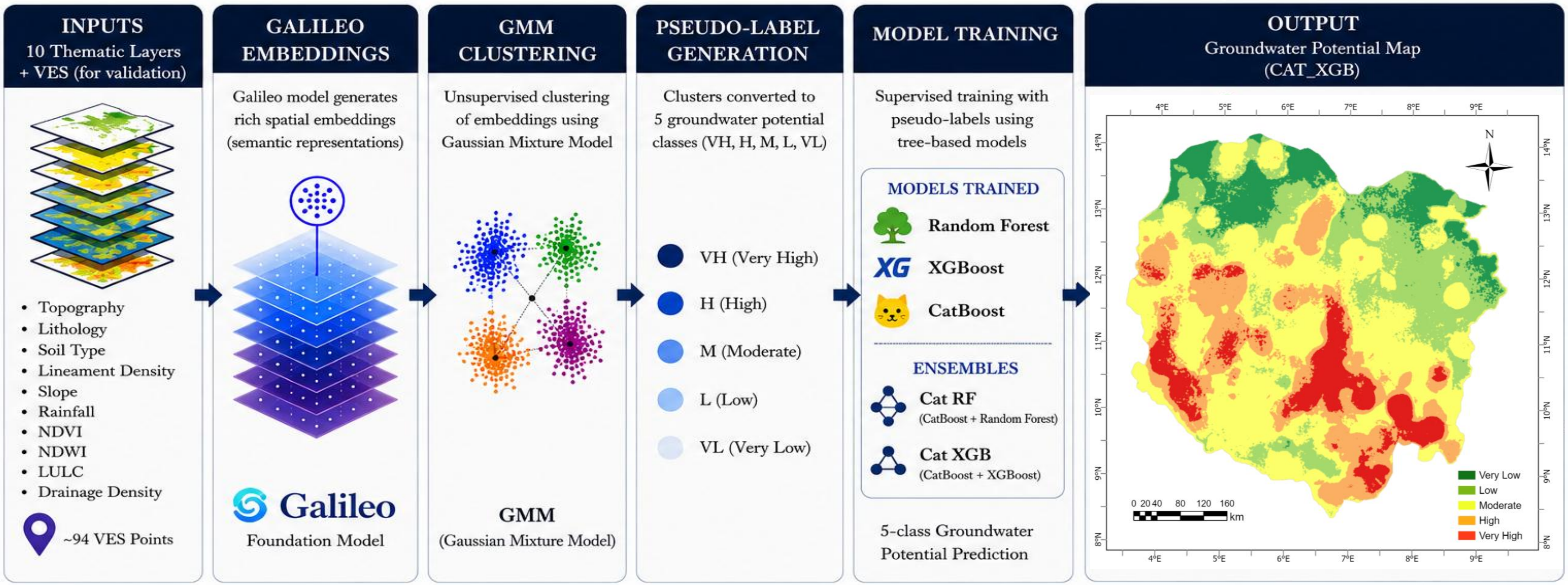
✓ Result is more optimistic/aggressive in predicting high potential areas

Recommended Models

CATB-XGB: Better Acc-95.4%, F1-95.5%

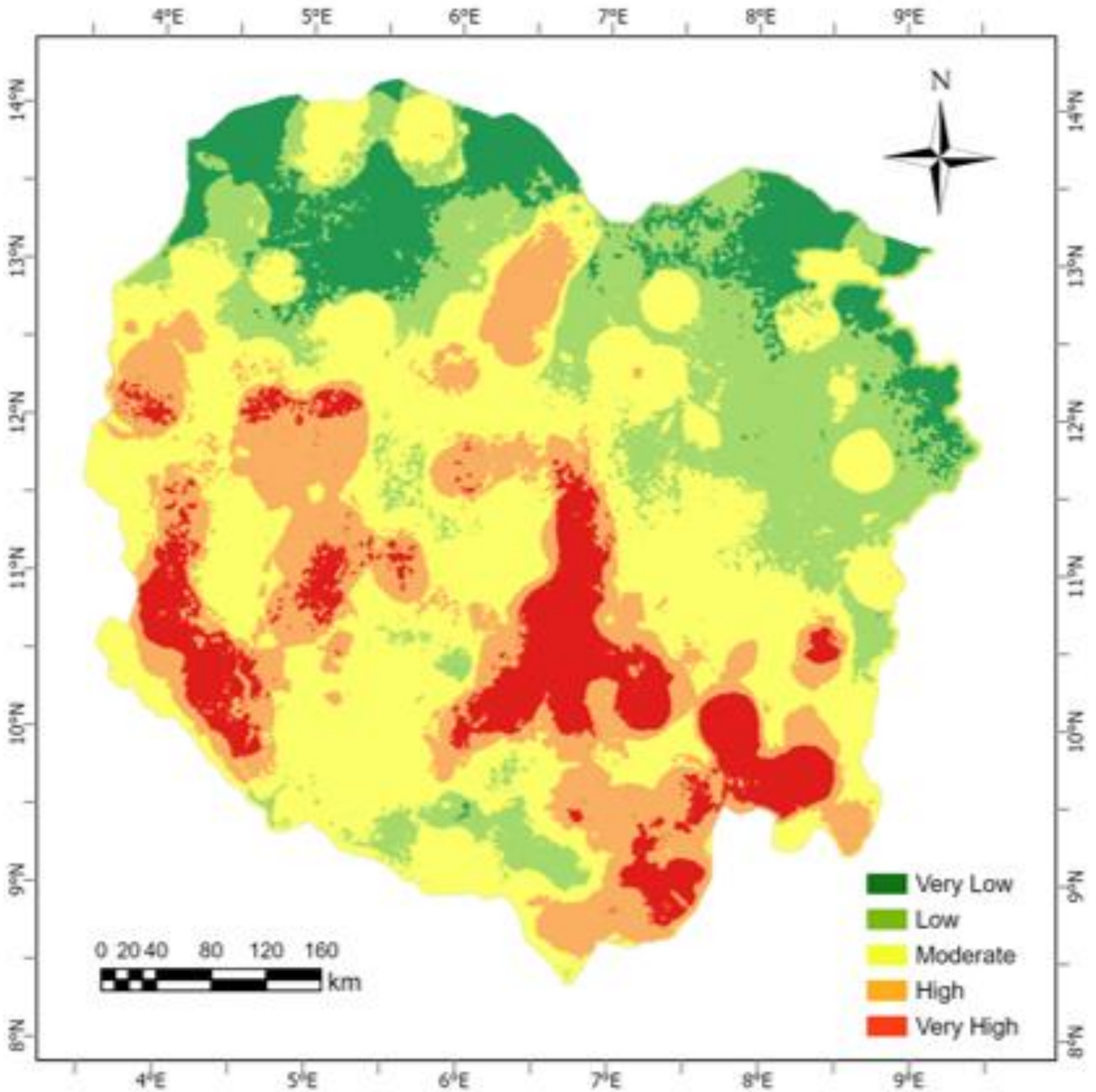
Acc:0.954, F1:0.955, Prec:0.955, Rec:0.954, Kappa:0.943

GALILEO MULTI-MODAL FRAMEWORK



Error Metrix: Acc:0.950, F1:0.948, Prec:0.950, Rec:0.946, Kappa:0.985

RESULTS —GMM (CAT_XGB) GROUNDWATER POTENTIAL MAP



Galileo Advantages

✓ It appears slightly smoother with more balanced yellow/moderate and green coverage, showing less extreme red patches

✓ The map has softer transitions and slightly larger contiguous moderate zones

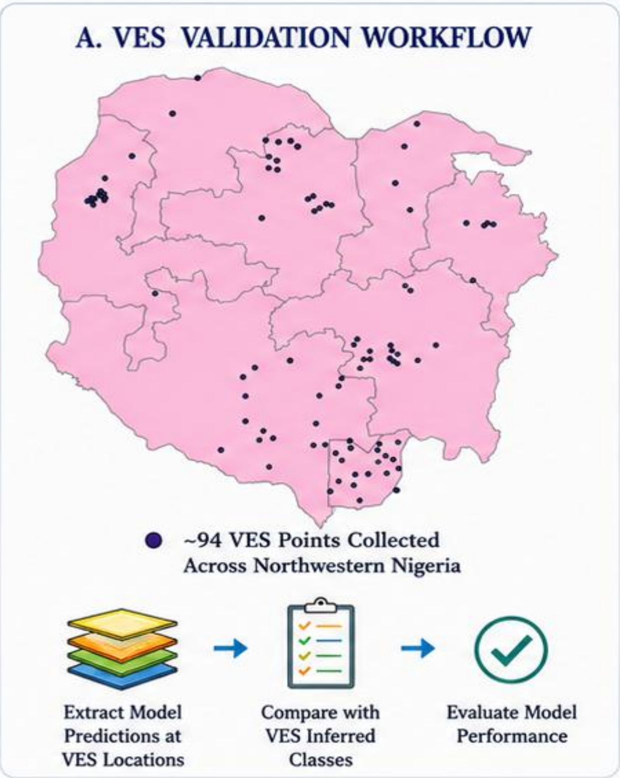
✓ It is a more conservative/smooth

Recommended Models

CATB-XGB: Better Acc-95.4%, F1-95.5%

Acc: 0.950, F1: 0.948, Prec: 0.950, Rec: 0.946, Kappa: 0.985

VES VALIDATION AND RESULTS



B. CONFUSION MATRIX (GALILEO - CAT_XGB)

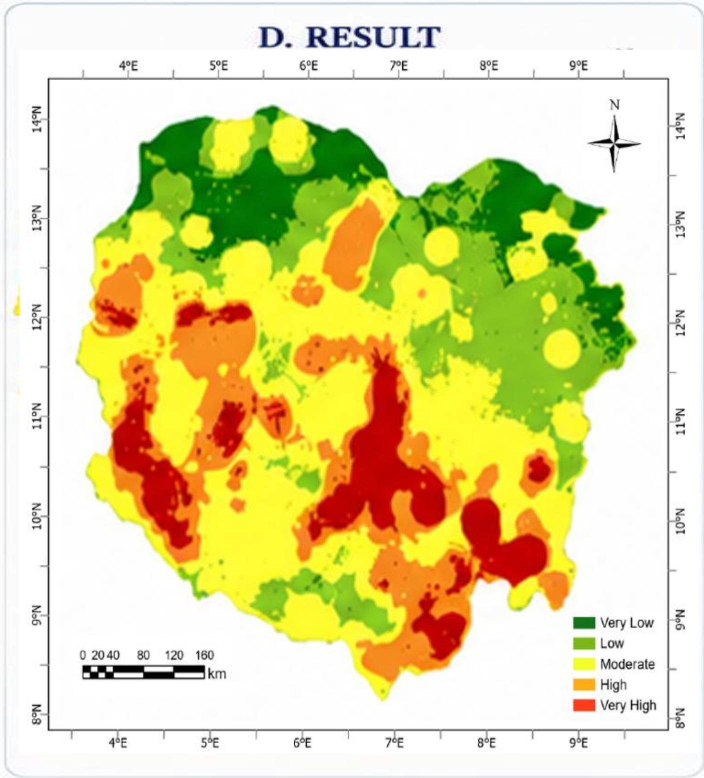
Predicted Class (columns) / True Class (rows)

	VH	H	M	L	VL
VH	45	3	0	0	0
H	2	40	2	0	0
M	1	3	39	2	1
L	0	1	4	41	1
VL	0	0	1	2	45

(Total VES Points = 94)

C. PERFORMANCE METRICS

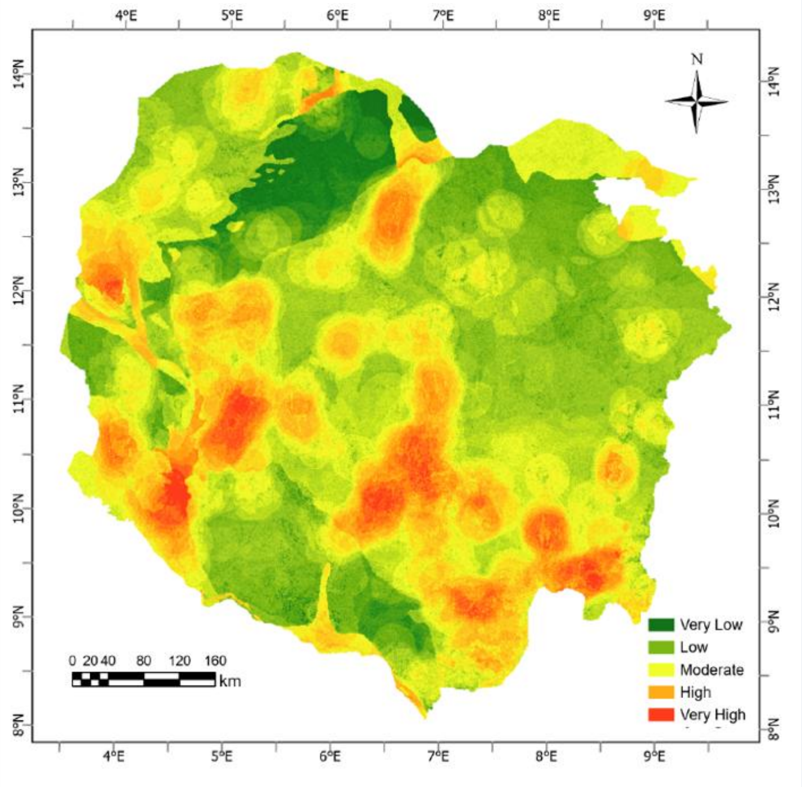
- Accuracy 0.90
- F1-Score (Macro) 0.89
- Kappa Coefficient 0.87
- ROC / AUC (OvR) 0.94



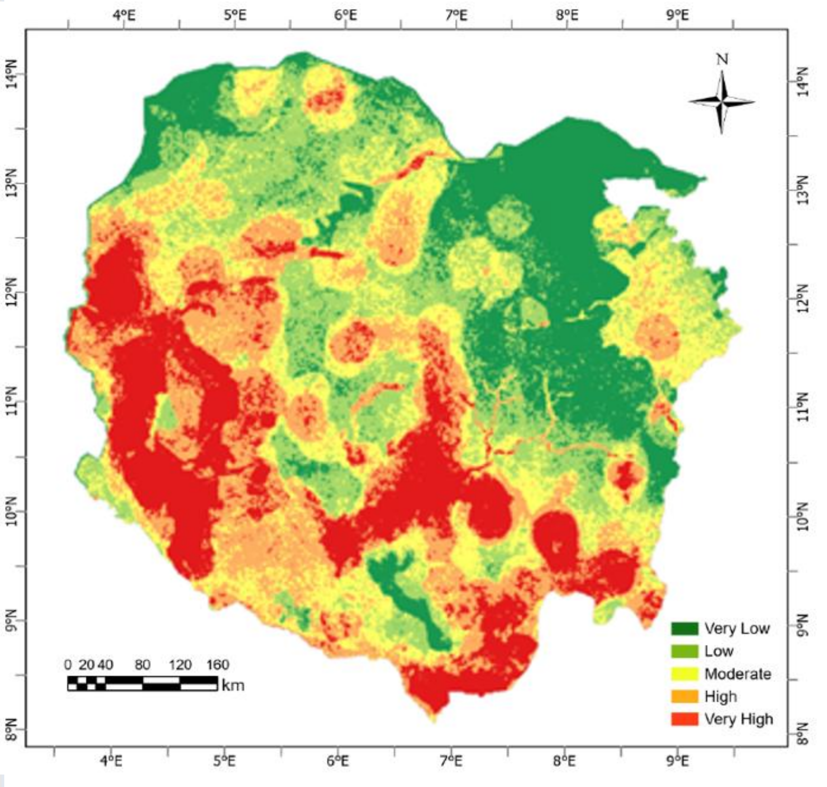
Error Metrix: Acc:0.988, F1:0.987, Prec:0.985, Rec:0.990, Kappa:0.938

COMPARISON OF GWP MAPS

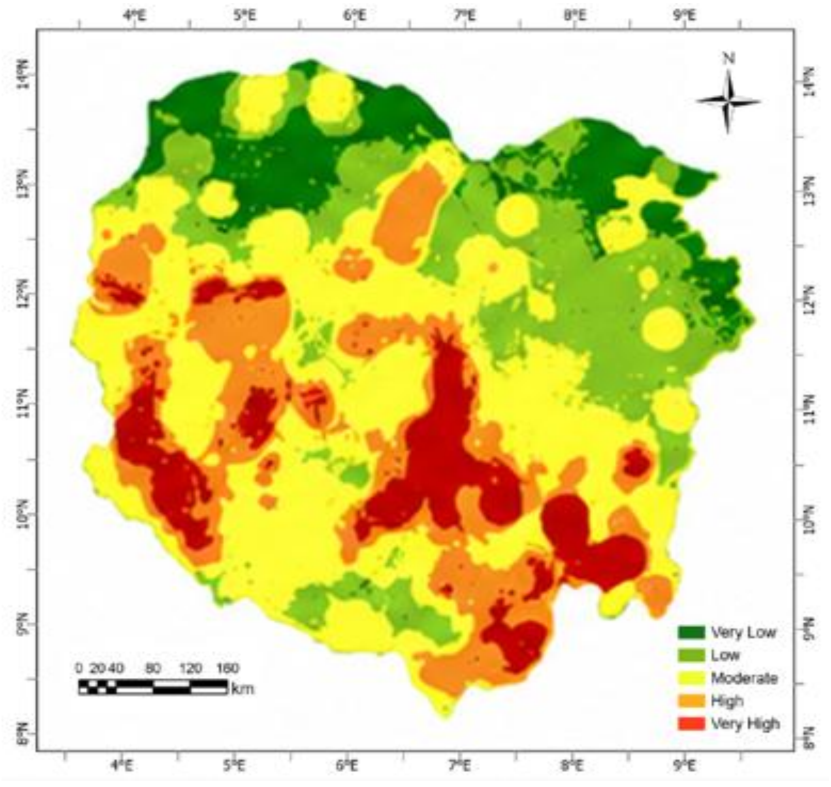
Method 1: AHP



Method 2: Raw ML (CATB)



Method 3: Galileo MM



Criterion	AHP	Raw ML (CATB)	Galileo ML (CAT_XGB)
Data requirement	GIS layers only	Layers + GMM- Derived labels	GIS layers + VES + GMM information
Spatial coherence	Moderate	Lower (noisy patches)	High (smooth zones, geologically coherent)
Ground truth needed	Expert judgement only (No VES)	VES used for model checking not training	VES used as geophysical guide & for validation
Role in this study	Policy oriented baseline map	ML Baseline reference	**Preferred product for detail GW mapping**

CONCLUSIONS

01

Multi-method validation

Integrating AHP, raw ML, and Galileo-ML produces a robust groundwater potential framework with cross-validation through VES borehole data.

02

AHP remains valuable

Saaty pairwise weighting with 10 GIS layers effectively delineates potential zones with expert-guided transparency and interpretability.

03

Galileo ML performance

The Galileo spatial embedding approach (CAT_XGB) produces smoother, geologically more realistic groundwater potential maps compared to pixel-based ML.

04

Key drivers identified

Lithology (31.4%) and lineament density (22.3%) dominate groundwater potential, guiding future borehole siting priorities in Northwestern Nigeria.

THE RESEARCH GROUP

Josef Osumeje, Schmidt AI in Science African Faculty Fellow @ MIDAS, U-M, Associate Professor of Geophysics, Ahmadu Bello University

Dani Jones (Mentor), Cooperative Institute of Great Lakes Research (CIGLR), SEAS U-M

Wanqi Li (Morli), Research Assistant, Expert in GIS and Remote Sensing, U-M

Sachet Sandeep Utekar, Research Assistant, Expert in AI and Data-Driven Modeling, U-M

Combining geophysics, remote sensing and spatial AI enables more reliable groundwater exploration in data-scarce regions.

Q&A

Thank you!